Aditya Morolia

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Education

B.Tech in Computer Science and MS by Research in Computational Natural Sciences — **IIIT Hyderabad**
July 2017–June 2023 — *India*

PhD, Computer Science — **Centre for Quantum Technologies (CQT), National University of Singapore (NUS)**
August 2024–Present — *Singapore*

Advisors: Divesh Aggarwal

Publications

- Divesh Aggarwal, Aditya Morolia, and Noah Stephens-Davidowitz. Faster provable lattice sieving with spherical codes. Cryptology ePrint Archive, Paper 2026/2388, 2026 (eprint)
- Divesh Aggarwal, Rishav Gupta, Aditya Morolia, and Chuanqi Zhang. Mind the Gap? Not for SVP Hardness under ETH! In *ICALP 2026*, volume 374 of *LIPIcs*, pages 8:1–8:24, 2026 (arXiv; ICALP26)
- Joao F. Doriguello, George Giapitzakis, Alessandro Luongo, and Aditya Morolia. On the practicality of quantum sieving algorithms for the shortest vector problem. In *PQCrypto 2026*, volume 16491 of *LNCS*, pages 3–36, 2026 (arxiv; QCTiP'25; PQCrypto'26)
- Shantanav Chakraborty, Aditya Morolia, and Anurudh Peduri. Quantum Regularized Least Squares. *Quantum*, 7:988, April 2023 (Quantum Journal)

Experience

Research Internship — **CIFRA, Bocconi University, Milano**
November 2025 - February 2026 — *Italy*
- Visiting Giulio Malavolta and Alon Rosen.

Research Assistant — **Centre for Quantum Technologies (CQT), National University of Singapore (NUS)**
August 2023 - July 2024 — *Singapore*

Research Assistant — **International Institute of Information Technology (IIIT) Hyderabad**
May 2019 - June 2023 — *Hyderabad, India*
- Advisors: Shantanav Chakraborty, Indranil Chakrabarty
- MS Thesis: Quantum Algorithms for Regularized Least Squares.

Mitacs Globalink Research Intern — **University of Calgary**
August 2021 - December 2021 — *Calgary, Canada*
- Advisor: Barry C Sanders

Engineering Analyst — **Goldman Sachs**
July 2022 - July 2023 — *GS Hyderabad, India*
- In the SecDB team, which designs and implements state of the art distributed database systems for financial trades.
- I was previously a summer analyst in the same team (2021).

Technical Staff — **AICrowd**
April 2020–July 2020 — *India*
- Developed a new **Django web app** AICrowd classroom; Developed a tool to simulate user behaviour on the website to study ranking systems.

Research Intern — **Indian School of Business (ISB), Mohali**
June 2019–August 2019 — *India*

Visiting Student — **Indian Statistical Institute (ISI) Kolkata**
May 2019 — *Kolkata, India*

Awards and Honours

- National Quantum Scholarship Scheme (NQSS) award (National Quantum Office, Singapore) *2024*
- Mitacs Globalink Undergraduate Research Fellowship *2021*
- IIIT Hyderabad Dean's Merit List Award for Academic Performance (top 20% of cohort) *Spring 2021*
- IIIT Hyderabad Dean's Merit List Award for Academic Performance (top 20% of cohort) *Monsoon 2020*
- AICrowd DroneRL (Reinforcement Learning) Competition winner *2020*
- Qualified JEE (Mains) (Among top 0.1% candidates) *2017*
- Qualified JEE (Advanced) (Among top 0.3% candidates) *2017*

Selected Talks

- **Expanding Frontiers of Cryptography: Assumptions & Applications, IIT Madras** — Faster Provable Lattice Sieving with Spherical Codes (Paper) Nov 2026
- **School of Computing, NUS** — Learning Quantum Hamiltonians from Ground States Oct 2026
- **School of Computing, NUS** — Quantum Monte Carlo Methods, Simplified Oct 2026
- **NUS Group Seminar** — Is there a Johnson-Lindenstrauss lemma for the ℓ_1 metric? Oct 2026
- **NUS Group Seminar** — Pseudorandom Error-Correcting Codes May 2026
- **NUS** — LoRA+: Efficient Low Rank Adaptation of Large Models (Slides) Apr 2026
- **CQT, NUS** — Gaussian Boson Sampling (Slides) Mar 2026
- **CQT CS Seminars** — Quantum Algorithms for Regularized Least Squares (Paper) Nov 2023
- **Singapore QML Seminars** — Quantum Signal Processing and Applications to Linear Systems (Paper) Nov 2023
- **Goldman Sachs, India Developers' Exchange** — Quantum Computing for Software Engineers Mar 2023
- **IIIT Hyderabad Theory Group** — Quantum Approximate Counting Aug 2020
  Started the Theory Reading Group's initiative "Seminar Saturdays" with a this talk.

Teaching And Mentorship

Teaching Assistant — NUS, Singapore

- **Optimization for Machine Learning**, 2026. Instructors: Patrick Rebentrost and Maciej Obremski.
- **Quantum Computing**, 2026. Instructor: Divesh Aggarwal.
- **Advanced Algorithms (Lattices, Integer Programming)**. Instructor: Divesh Aggarwal. lattice lecture notes.

Instructor, QCamp 2026 — Centre for Quantum Technologies, Singapore

- Taught complex numbers and linear algebra to pre-university students. Built interactive TypeScript/Canvas visualizations and companion slides to explain the mathematics underlying quantum computing. Interactive tools; Slides; Code

Teaching Assistant — IIIT Hyderabad, India

- **Linear Algebra**, 2020 (Undergraduate and Graduate)
- **Introduction to Quantum Information and Computation**, 2021 (Elective)
- **Automata Theory**, 2021 (Undergraduate)
- **Quantum Algorithms**, 2022 (Elective)

Guest Lecture for Linear Algebra Course: Applications of Linear Algebra · 2022 — **IIIT Hyderabad, India**

Mentor, Singapore Quantum Youth Collective (SQYC) · 2026 — **Singapore**

Mentor, Women in STEM Research (WISR) · 2023 — **Remote**

Selected Projects

- **Distributed MapReduce** (C++, Boost, MPI) — Worker-failure handling and load balancing. GitHub
- **C- - Compiler** (C++, Antlr) — Compiler for a toy language inspired by C and Decaf. GitHub
- **C Shell** (C, Linux) — Toy shell. GitHub
- **Graph Neural Network** (PyTorch, Cirq) — ParticleNet for particle data classification. GitHub
- **QSim** (TypeScript, CSS, SVG, Vite) — Offline circuit simulator, puzzles and synchronized displays. GitHub
- **Lattice Visualizer** (TypeScript, Canvas, Vite) — Lattices are beautiful mathematical objects. Let's try to see them! GitHub
- **Automated Bidding Agent** (Java) — Budget-constrained bidding for targeted ad space.
- **Molecular Dynamics Simulation** (Python) — MCMC with periodic boundaries and minimum-image convention.